

How Far Does Treatment Go?

A Global Analysis of ART Treatment Coverage and HIV Mortality Patterns

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Introduction

HIV/AIDS remains one of the most significant global public health challenges, affecting millions of individuals worldwide and continuing to place a substantial burden on healthcare systems. According to the World Health Organization, approximately 39 million people are currently living with HIV globally, and HIV-related illness continue to account for hundreds of thousands of deaths each year (World Health Organization, 2023). Despite major advances in treatment and prevention, the burden of HIV/AIDS is not evenly distributed across countries, with some regions experiencing substantially higher mortality rates than others.

HIV/AIDS outcomes at a global scale are closely linked to access to antiretroviral therapy (ART) and other structural factors that vary across countries. ART suppresses the replication of the virus, allowing individuals living with HIV to maintain healthier immune systems and significantly lowering the risk of progression to AIDS and death (World Health Organization, 2023). As a result, increasing access to ART has become a central focus of global health initiatives, but ART varies widely across countries due to differences in healthcare infrastructure, economic resources, and public health policy.

Because of this variation, it is important to understand whether higher ART coverage is actually associated with lower HIV/AIDS death counts when comparing countries. In addition, countries with larger populations of people living with HIV are naturally expected to have higher numbers of deaths, regardless of treatment coverage. This makes it necessary to account for population size when evaluating the relationship between ART coverage and mortality. HIV/AIDS death counts are also highly skewed across countries, with a small number of countries contributing disproportionately large values. These characteristics demonstrate the need for both adjusted models and transformations.

In this study, we examine the association between ART coverage and HIV/AIDS deaths across countries using country-level data. We explore both unadjusted and adjusted relationships, incorporating the number of individuals living with HIV as a control variable and applying log transformations to better capture the relationship in the data. This leads to our primary research question:

Is higher antiretroviral therapy (ART) coverage associated with lower HIV/AIDS death counts across countries?

Data

The dataset used in this analysis was constructed by merging three country-level datasets containing HIV/AIDS indicators: HIV/AIDS deaths, the number of people living with HIV, and antiretroviral therapy (ART) coverage among people living with HIV. The original data were accessed from a GitHub repository (<https://github.com/srivi15/HIV-AIDS---Data-Analysis>) and are derived from publicly available sources, including the World Health Organization (WHO), UNESCO, and Kaggle.

Prior to merging, each dataset was cleaned and standardized to ensure consistency across variables. Variable names were reformatted, and the data were aggregated at the country level. For both the HIV/AIDS deaths

and HIV population datasets, median estimates were used to represent each country. The ART dataset was also cleaned to only the relevant variable measuring the percentage of individuals living with HIV who are receiving treatment.

The final dataset contains 170 countries, with each observation representing a single country. The primary variables used in the analysis are HIV/AIDS deaths (deaths), number of people living with HIV (hiv_population), and ART coverage (art_coverage). All variables are continuous numerical estimates.

```
##          country      deaths hiv_population  art_coverage
##             0          37          32          34

##          country      deaths hiv_population  art_coverage
##    0.0000000    0.2176471    0.1882353    0.2000000
```

Missing values are present in all main variables: HIV/AIDS deaths (37 missing, ~21.8%), HIV population (32 missing, ~18.8%), and ART coverage (34 missing, ~20%). These missing values likely reflect the real-world issue of incomplete reporting in global health surveillance systems. For the analysis, missing data were handled using listwise deletion, meaning observations with missing values were automatically excluded when fitting each model.

Methods

To examine the relationship between ART coverage and HIV/AIDS deaths across countries, we conducted a series of statistical analyses. We begin with a correlation test to assess the unadjusted relationship between ART coverage and deaths. We then fit a sequence of regression models, including a simple linear regression, a multiple linear regression controlling for the number of people living with HIV, and a log-transformed model to account for skewness in the data. Each model builds on limitations of the previous one and is interpreted in the sections below.

The correlation analysis provides an initial measure of strength and direction of the linear association between ART coverage and HIV/AIDS deaths, but it does not account for differences across countries. To address this, a simple linear regression model was first fit with HIV/AIDS deaths as the response variable and ART coverage as the predictor. This serves as a baseline model for the relationship without any adjustments.

A multiple linear regression model was then used to include both ART coverage and the number of individuals living with HIV, allowing us to control for differences in HIV burden across countries. This is important because countries with larger HIV-positive populations are expected to have higher numbers of deaths regardless of ART coverage.

Finally, because HIV/AIDS death counts are highly skewed, a log-transformed regression model was used, in which the natural logarithm of deaths and HIV population were included. Model assumptions were assessed using diagnostic plots, including residuals versus fitted values and histograms of residuals, to evaluate linearity, constant variance, and normality.

Results

Correlational Analysis

```
## [1] 0.02297943
```

The correlation between ART coverage and HIV/AIDS was very weak ($r = 0.023$), indicating little to no linear relationship when examined alone. However, this result may be misleading because it does not account for differences in the number of people living with HIV across countries, which can strongly influence death counts. This suggests that a controlled regression analysis is necessary.

Simple Linear Regression

```
##
## Call:
## lm(formula = deaths ~ art_coverage, data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4934  -4400  -3800  -1538   66292
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3960.15    2697.69   1.468   0.145
## art_coverage    12.07     46.22   0.261   0.794
##
## Residual standard error: 10580 on 129 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.0005281, Adjusted R-squared: -0.00722
## F-statistic: 0.06815 on 1 and 129 DF, p-value: 0.7945
```

A simple linear regression was conducted to examine the relationship between ART coverage and HIV/AIDS deaths. The results showed no significant relationship between ART coverage and deaths ($\beta = 12.07$, $p = 0.794$). Additionally, the model explained virtually none of the variation in deaths ($R^2 = 0.0005$), indicating that ART coverage alone is not a meaningful predictor of HIV/AIDS mortality. This suggests that other factors, such as the size of the HIV-positive population, may play a more important role.

Multiple Linear Regression

```
##
## Call:
## lm(formula = deaths ~ art_coverage + hiv_population, data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23510.6  -1746.6  -1210.4   -276.6   28179.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.021e+03  1.347e+03   2.985  0.0034 **
## art_coverage  -4.055e+01  2.323e+01  -1.745  0.0833 .
## hiv_population  1.208e-02  6.121e-04  19.732 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5285 on 128 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.7527, Adjusted R-squared:  0.7489
## F-statistic: 194.8 on 2 and 128 DF, p-value: < 2.2e-16
```

A multiple linear regression model was used to examine the relationship between ART coverage and HIV/AIDS deaths while controlling for the number of people living with HIV. The results showed that ART coverage had a negative association with deaths ($\beta = -40.55$, $p = 0.083$), suggesting a potential negative relationship between ART coverage and HIV/AIDS deaths. Although this relationship was not statistically significant at the 0.05 level, it indicates a possible downward trend.

In contrast, the number of people living with HIV was a highly significant predictor of deaths ($\beta = 0.01208$,

$p = 2 * 10^{-16}$), indicating that countries with larger HIV-positive populations experience substantially more deaths.

The model demonstrated strong explanatory power ($R^2 = 0.753$), meaning that approximately 75% of the variation in HIV/AIDS deaths across countries is explained by these variables.

All in all, while ART coverage alone did not appear to influence deaths in the simple model, the controlled regression suggests that increased ART coverage is associated with reduced HIV/AIDS mortality when accounting for differences in HIV population size. This is consistent with the hypothesis that higher ART coverage is associated with lower HIV-related deaths globally.

Log-Transformed Model

```
##
## Call:
## lm(formula = log(deaths) ~ art_coverage + log(hiv_population),
##     data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.61043 -0.38433  0.01353  0.34033  2.16313
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.662964    0.275172  -2.409   0.0174 *
## art_coverage   -0.014334    0.002538  -5.649 9.96e-08 ***
## log(hiv_population)  0.799767    0.023634  33.840 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5792 on 128 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.9001, Adjusted R-squared:  0.8986
## F-statistic: 576.8 on 2 and 128 DF,  p-value: < 2.2e-16
```

A log-transformed multiple regression model was used to better account for skewness in the data and examine the relationship between ART coverage and HIV/AIDS deaths. The model takes the form:

$$\log(\text{deaths}) = \beta_0 + \beta_1(\text{art_coverage}) + \beta_2 \log(\text{hiv_population}) + \epsilon$$

The results showed that ART coverage had a statistically significant negative association with $\log(\text{HIV/AIDS deaths})$ ($\beta = -0.0143$, $p = 9.96e-08$), after controlling for $\log(\text{HIV population size})$. This means that for each 1 percentage point increase in ART coverage, a country's expected HIV/AIDS deaths are multiplied by a factor of $e^{(-0.0143)} = 0.986$. This is about a 1.4% decrease in deaths per additional percentage point of ART coverage.

$\log(\text{HIV population size})$ was also highly significant ($\beta = 0.800$, $p < 2e-16$). This confirms that population size remains the main contributor to death counts.

This model explained 90.0% of the variation in $\log(\text{HIV/AIDS deaths})$ ($R^2 = 0.900$), a substantial improvement over both the simple ($R^2 = 0.0005$) and controlled ($R^2 = 0.753$) untransformed models, suggesting the log transformation better captures the underlying relationships in the data.

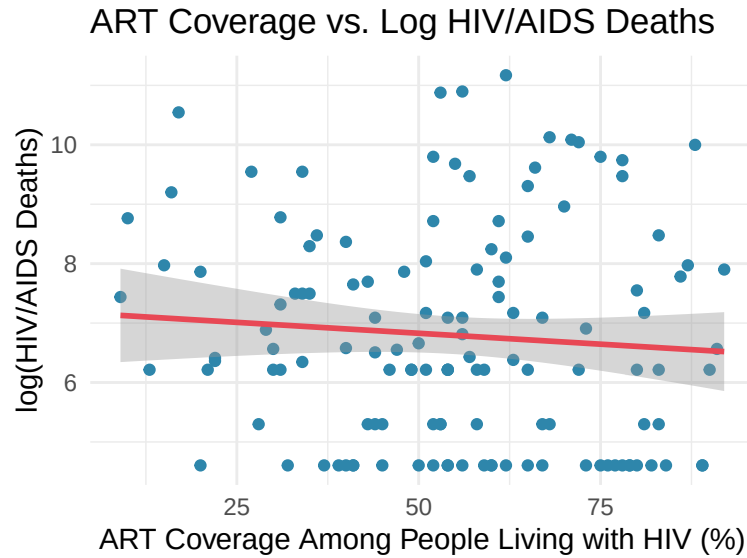


Figure 1 displays the relationship between ART coverage and $\log(\text{HIV/AIDS deaths})$ across countries, with a fitted regression line from Model 3 and 95% confidence band. The negative slope visually shows that higher ART coverage is associated with lower HIV/AIDS mortality, while controlling for HIV population size.

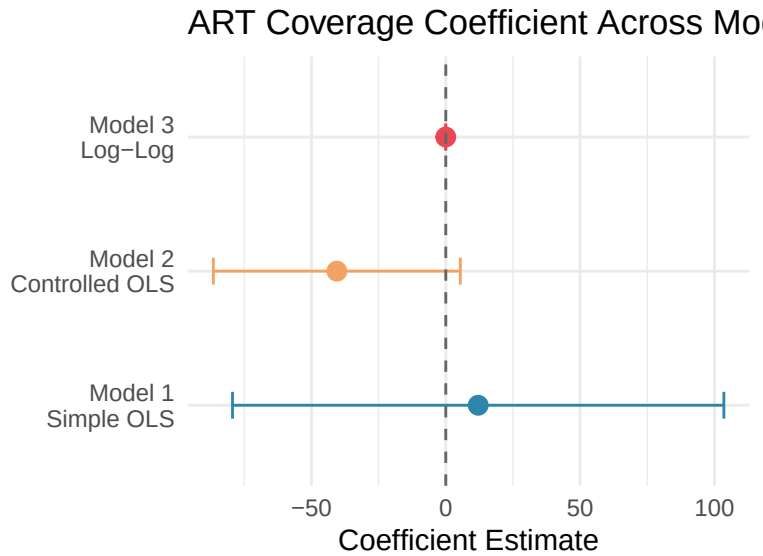


Figure 2 compares the ART coverage coefficient estimate across all three model specifications with 95% confidence intervals. In Models 1 and 2, the confidence interval crosses zero, indicating no statistically significant effect. Only in Model 3, after log-transforming both the outcome and HIV population size, does the estimate become significant, supporting the use of the log-log specification.

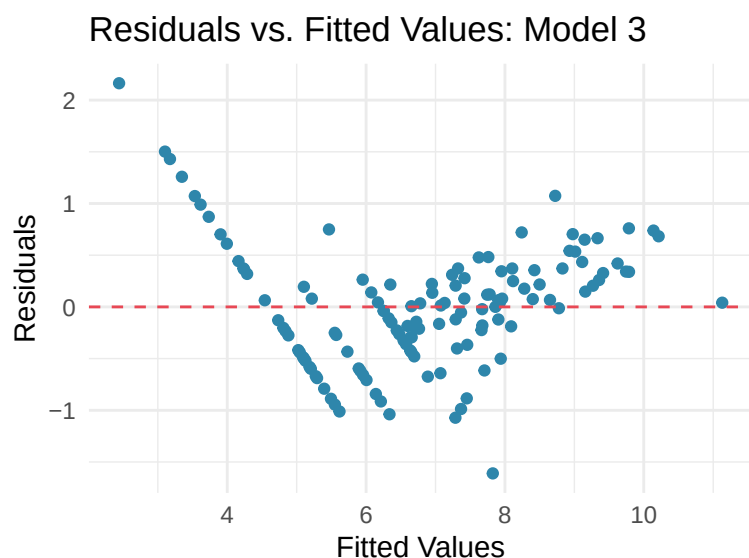


Figure 3 shows residuals scattered approximately randomly around zero with no strong systematic pattern, suggesting the linearity and constant variance assumptions of Model 3 are reasonably satisfied.

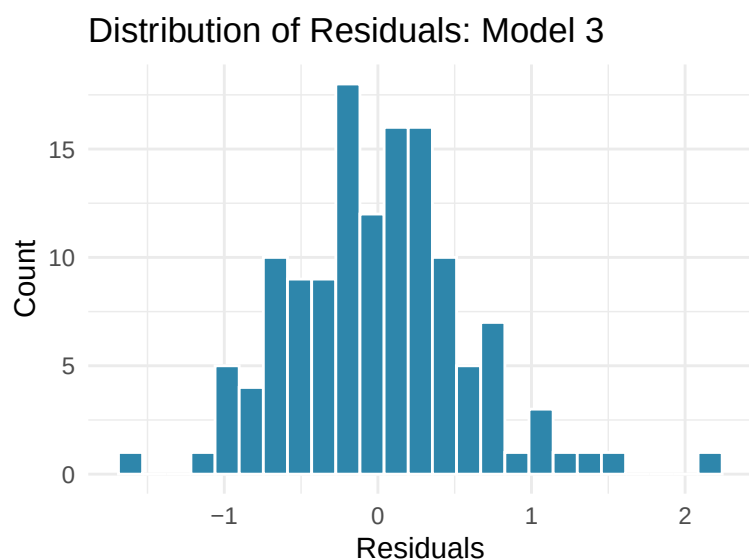


Figure 4 shows the distribution of residuals from Model 3. The approximately bell-shaped, symmetric distribution centered near zero suggests the normality assumption is reasonably met, supporting the validity of the model's inference.

Discussion

This analysis examined the relationship between ART coverage and HIV/AIDS deaths across countries using multiple statistical models. The results indicate that the relationship between ART coverage and mortality depends strongly on model specification. In the initial correlation and simple regression models, ART coverage showed little to no association with HIV/AIDS deaths. However, these results may be misleading because they do not account for differences in the size of the HIV-positive population across countries.

After controlling for the number of people living with HIV, ART coverage showed a negative association with HIV/AIDS deaths, suggesting that higher treatment coverage is associated with fewer deaths. Although this

relationship was not statistically significant at the 0.05 level in the multiple linear regression model, the direction and magnitude of the effect suggest a potential downward trend. The log-transformed model provided the strongest evidence of a relationship, showing a statistically significant negative association between ART coverage and HIV/AIDS deaths after adjusting for HIV population size.

These results suggest that model specification plays an important role in identifying the relationship between ART coverage and HIV/AIDS outcomes. In particular, the improved fit of the log-transformed model indicates that accounting for skewness in both deaths and HIV population provides a more appropriate representation of the data structure.

Despite these findings, several limitations should be considered. The analysis is based on observational, country-level data, which limits the ability to make causal conclusions about the relationship between ART coverage and HIV/AIDS deaths. The presence of missing data across key variables may introduce bias and affect the reliability of estimates. The models also do not account for other potentially important factors, such as healthcare infrastructure, economic conditions, or access to prevention and testing services, which may influence HIV/AIDS outcomes across countries. Future work could improve this analysis by incorporating additional country-level variables and using longitudinal data to examine how changes in ART coverage are associated with changes in mortality over time. These extensions would provide a more complete understanding of the drivers of HIV/AIDS outcomes.

Conclusion

In conclusion, this study examined the relationship between ART coverage and HIV/AIDS deaths across countries using correlation analysis and multiple regression models. While initial analyses suggested little to no association between ART coverage and mortality, more refined models that accounted for HIV population size and data skewness revealed a significant negative relationship. In particular, the log-transformed model provided the strongest evidence that higher ART coverage is associated with lower HIV/AIDS deaths. These findings demonstrate the importance of expanding access to antiretroviral therapy as a key strategy for reducing HIV/AIDS mortality at a global scale.

References

- Srividhya R. (n.d.). HIV AIDS – Data Analysis and Visualization. GitHub. <https://github.com/srivi15/HIV-AIDS---Data-Analysis>
- World Health Organization. (2023). HIV/AIDS. <https://www.who.int/news-room/fact-sheets/detail/hiv-aids>